BME 105: Genetics in the Genomics Era

Lecture 16: Genome-Wide Association Studies (GWAS) and Quantitative Trait Locus (QTL) analyses

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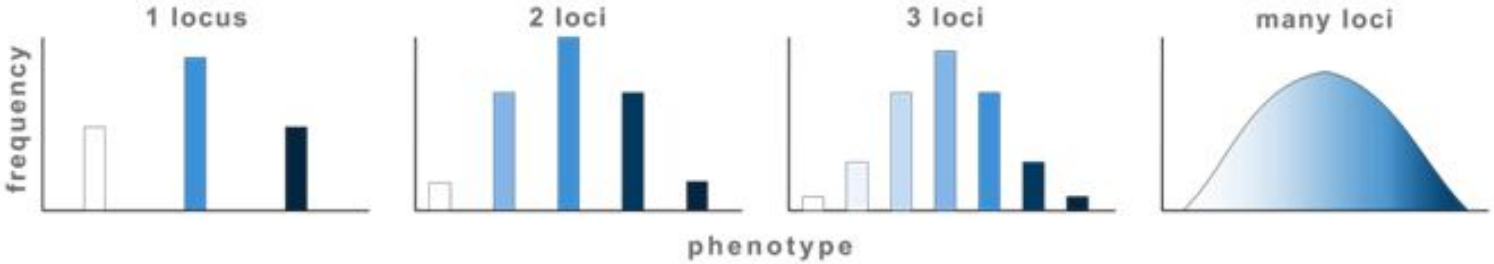
Course website on canvas.ucsc.edu

Lecture 16 Learning Objectives

- Understand how gene-gene interactions and allelic diversity increases phenotypic diversity
  - Continuous traits arise from this phenotypic complexity
  - The impact of each locus on the trait decreases as the number of encoding genes increases
- Be able to distinguish QTL from GWAS studies
  - Fine-scale mapping vs coarse mapping with high amounts of linkage
- Understand how recombination enables independent association testing between alleles and phenotypes
- Understand how linkage means that a QTL/associated SNP may not be causal
  - However, linkage does help with detection - the causal SNP is in the linked haplotype block
- Begin to understand how expression (and other epigenetic readouts) can be integrated with GWAS and QTL mapping.

Continuous complex traits manifest from allele-gene interactions

Incompletely dominant alleles; additive interactions



	A	a
A	AA	Aa
a	Aa	aa

	AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	AAbb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

	ABC	ABc	AbC	Abc	aBC	aBc	abC	abc
ABC	AABBCC	AABBCc	AABbCC	AABbCc	AaBBCC	AaBBCc	AaBbCC	AaBbCc
ABc	AABBCc	AABBcc	AABbCc	AABbcc	AaBBCc	AaBBcc	AaBbCc	AaBbcc
AbC	AABbCC	AABbCc	AAbbCC	AAbbCc	AaBbCC	AaBbCc	AabbCC	AabbCc
Abc	AABbCc	AABbcc	AAbbCc	AAbbcc	AaBbCc	AaBbcc	AabbCc	Aabbcc
aBC	AaBBCC	AaBBCc	AaBbCC	AaBbCc	aaBBCC	aaBBCc	aaBbCC	aaBbCc
aBc	AaBBCc	AaBBcc	AaBbCc	AaBbcc	aaBBCc	aaBBcc	aaBbCc	aaBbcc
abC	AaBbCC	AaBbCc	AabbCC	AabbCc	aaBbCC	aaBbCc	aabbCC	aabbCc
abc	AaBbCc	AaBbcc	AabbCc	Aabbcc	aaBbCc	aaBbcc	aabbCc	aabbcc

>3 genes, individual <u>classes may become indistinguishable</u> from each other; particularly when environmental effects are included

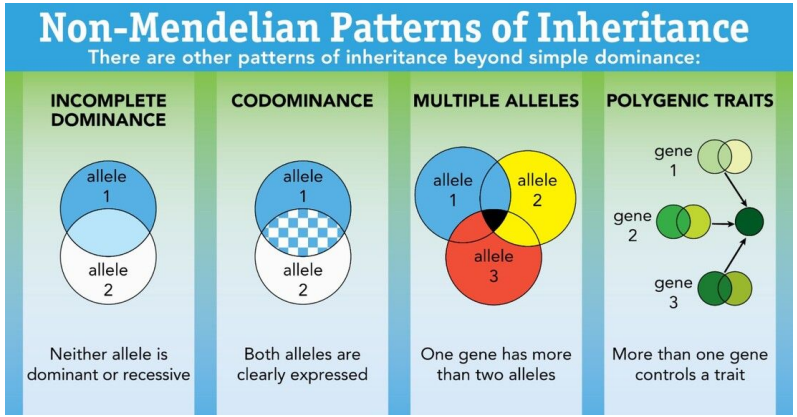
Complex traits are the most difficult to map to genotype

~ multigenic / polygenic traits

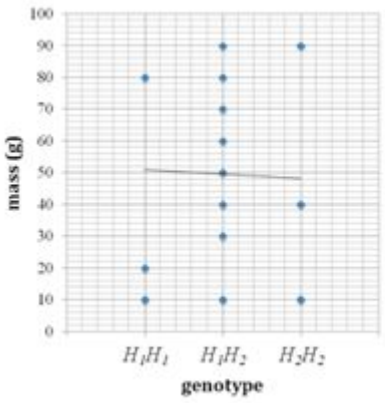
Continuous distribution of values

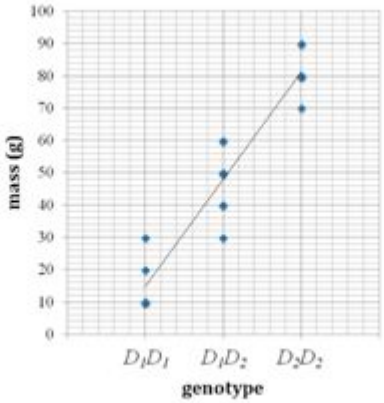
Environment may influence trait

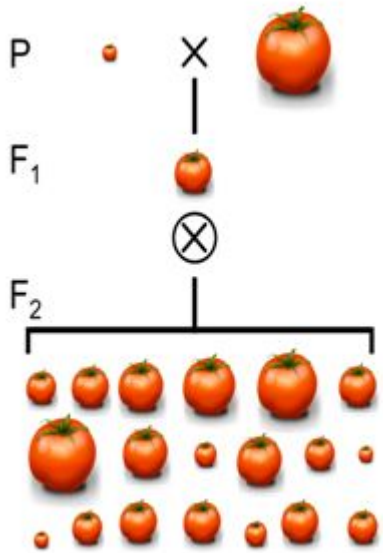
As a result of the subtle functional effect(s), in complex traits there is only <u>weak genotype-phenotype correlation</u>, with <u>low effect sizes</u> for the individually associated genetic variants.

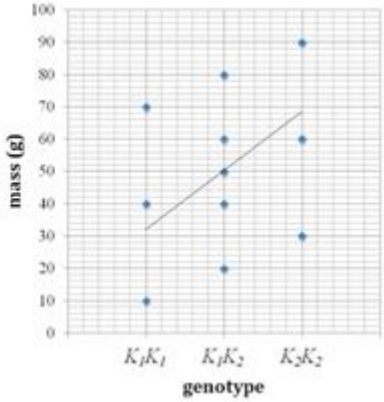


Incomplete/co-dominance and additive interactions increase phenotypic space, leading to continuous complex trait phenotypes









As a result of the subtle functional effect(s), in complex traits there is only <u>weak genotype-phenotype correlation</u>, with <u>low effect sizes</u> for the individually associated genetic variants.

	genotype	fruit mass
P	$A_1A_1B_1B_1C_1C_1D_1D_1E_1E_1F_1F_1G_1G_1H_1H_1I_1I_1K_1K_1$	10g
P	$A_2A_2B_2B_2C_2C_2D_2D_2E_2E_2F_2F_2G_2G_2H_2H_2I_2I_2K_2K_2$	90g
F_1	$A_1A_2B_1B_2C_1C_2D_1D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_2K_1K_2$	50g
F_2 #001	$A_1A_1B_1B_2C_1C_1D_2D_2E_1E_2F_1F_2G_1G_2H_1H_1I_1I_2K_1K_2$	80g
F_2 #002	$A_1A_2B_1B_2C_1C_2D_1D_1E_1E_2F_1F_2G_2G_2H_1H_2I_2I_2K_1K_1$	10g
F_2 #003	$A_2A_2B_1B_2C_2C_2D_1D_2E_1E_2F_1F_2G_1G_1H_1H_2I_1I_2K_1K_2$	50g
F_2 #004	$A_1A_2B_1B_2C_1C_2D_1D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_2K_2K_2$	60g
F_2 #005	$A_1A_2B_1B_1C_1C_2D_2D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_1K_2K_2$	90g
F_2 #006	$A_1A_2B_2B_2C_1C_2D_1D_2E_1E_2F_1F_2G_2G_2H_1H_2I_1I_2K_1K_2$	60g
F_2 #007	$A_2A_2B_1B_1C_1C_2D_2D_2E_1E_2F_1F_2G_1G_1H_1H_2I_1I_1K_1K_2$	80g
F_2 #008	$A_1A_1B_1B_2C_1C_2D_1D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_2K_1K_2$	50g
F_2 #009	$A_1A_2B_1B_2C_2C_2D_1D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_2K_1K_2$	50g
F_2 #010	$A_1A_2B_1B_2C_1C_2D_1D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_2K_2K_2$	30g
F_2 #011	$A_1A_2B_1B_2C_1C_2D_2D_2E_1E_1F_1F_2G_1G_2H_1H_2I_1I_2K_1K_2$	80g
F_2 #012	$A_1A_1B_1B_2C_1C_2D_1D_1E_1E_2F_2F_2G_1G_2H_1H_2I_1I_2K_2K_2$	30g
F_2 #013	$A_2A_2B_1B_1C_1C_2D_1D_1E_1E_2F_1F_1G_1G_2H_2H_2I_1I_1K_1K_1$	10g
F_2 #014	$A_2A_2B_1B_1C_1C_1D_2D_2E_1E_2F_1F_2G_1G_2H_1H_2I_1I_1K_1K_1$	70g
F_2 #015	$A_2A_2B_2B_2C_1C_2D_1D_2E_1E_2F_2F_2G_1G_2H_1H_1I_2I_2K_1K_2$	40g
F_2 #016	$A_1A_2B_2B_2C_1C_2D_1D_1E_1E_2F_1F_1G_2G_2H_1H_1I_1I_2K_1K_1$	10g
F_2 #017	$A_1A_2B_2B_2C_1C_2D_2D_2E_2E_2F_1F_1G_2G_2H_1H_2I_1I_2K_2K_2$	90g
F_2 #018	$A_1A_2B_2B_2C_1C_2D_1D_2E_1E_2F_1F_1G_2G_2H_1H_2I_1I_2K_1K_1$	40g
F_2 #019	$A_1A_1B_1B_2C_1C_2D_1D_1E_1E_2F_2F_2G_1G_1H_1H_1I_1I_2K_1K_2$	20g
F_2 #100	$A_1A_1B_1B_2C_1C_2D_2D_2E_1E_2F_1F_2G_2G_2H_1H_2I_2I_2K_1K_2$	80g

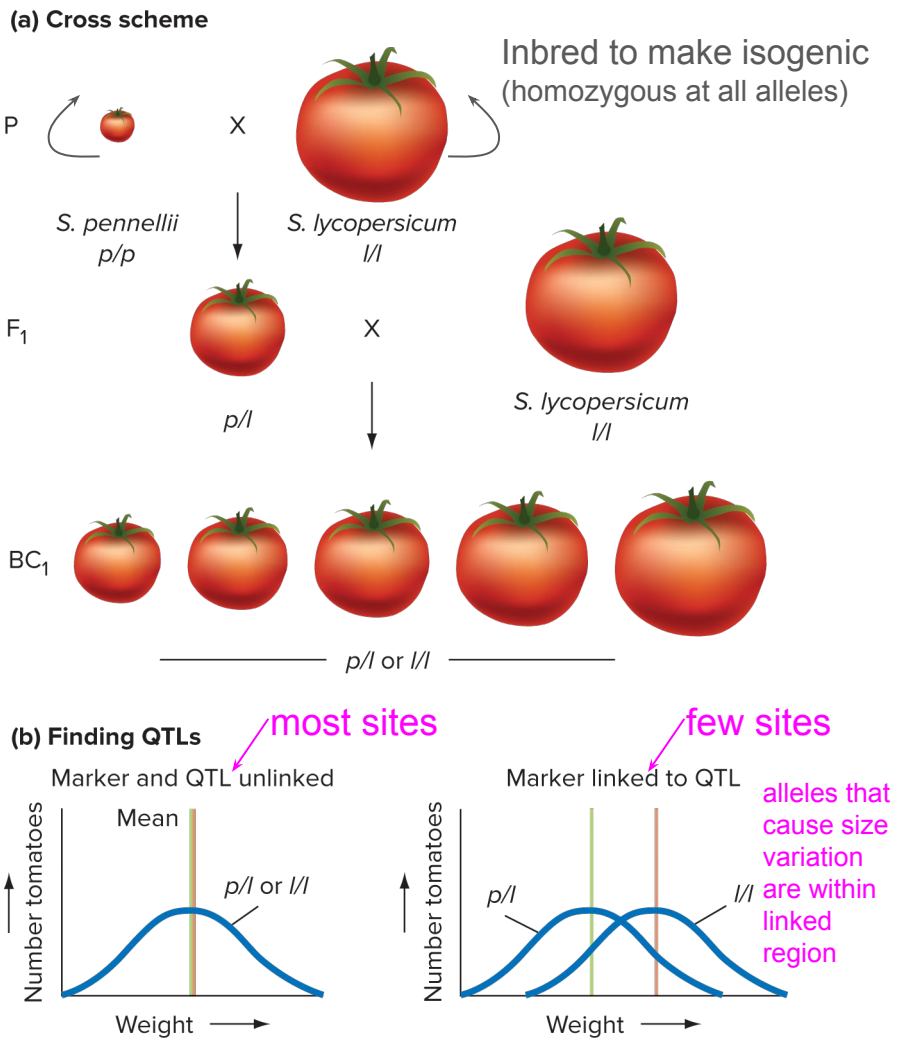
How can all of the loci that affect a trait be mapped?

- Several genes control tomato size
  - Small vs large parents have different alleles across many genes

    → BC_1 generation exhibits a wide range of sizes

    ⇒ researchers weighed these and tested for association between each allele and weight

Note the reliance on recombination to produce individuals with different genetic compositions that will give rise to different phenotypes.



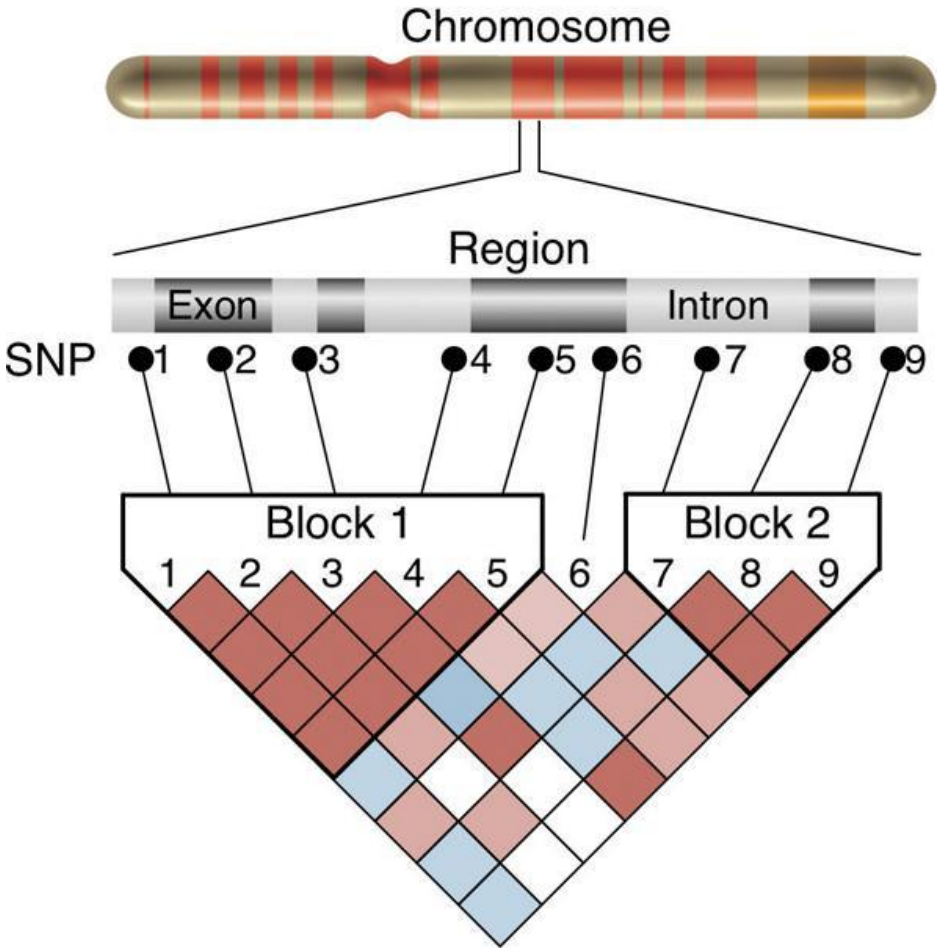
Linkage disequilibrium represented as a triangle diagram

$1x10^{-8}$ crossover events per generation between adjacent bp

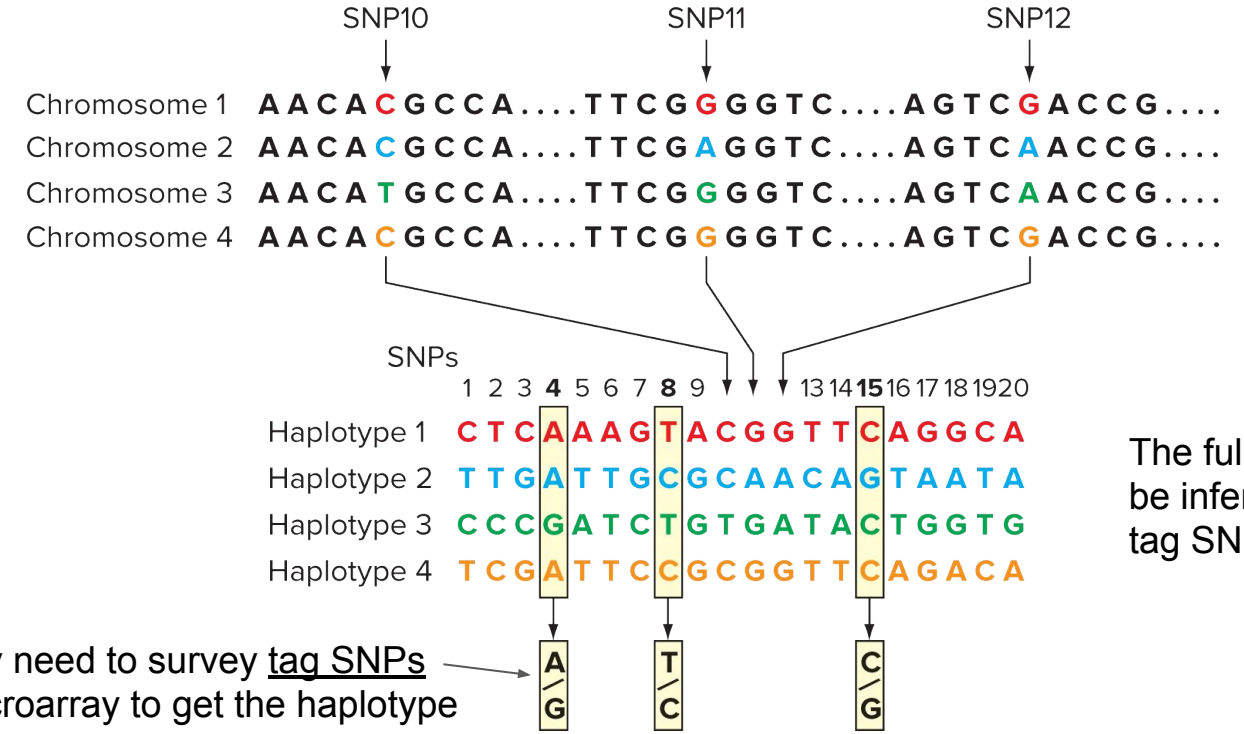
Block color indicates the strength of the statistical correlation (LD: linkage disequilibrium) for pairwise comparisons of the indicated SNPs

Darker color indicates stronger correlation

What is likely occurring in between Block 1 and Block 2?



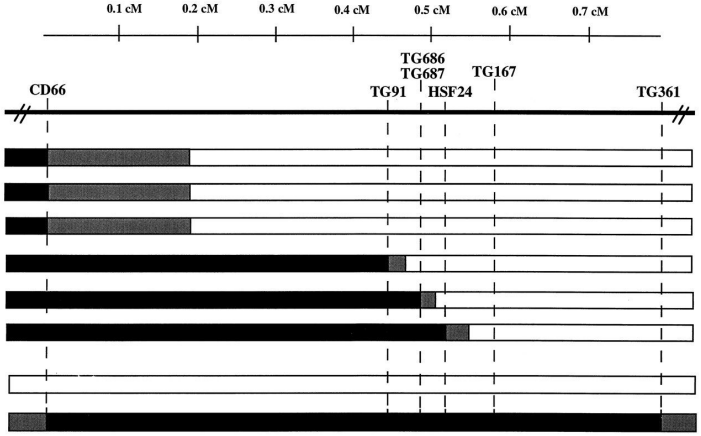
LD makes it cheaper to identify SNPs associated with complex traits by using SNP microarrays



The full haplotype can be inferred from the tag SNPs

You only need to survey <u>tag SNPs</u> on a microarray to get the haplotype

How can all of the loci that affect a trait be fine mapped?



Plant Recombinant id #	Average 10 Fruit Weight (g) CA	NY
3	NA	62.1 [b]
11	69.3 [b]	NA
12	63.3 [b]	NA
31	NA	63.0 [b]
33	NA	50.5 [a]
34	41.1 [a]	44.1 [a]
Controls		
M82-1-8	72.4	71.9
NIL 939-2	49.9	49.6

= Homozygous *Lycopersicon pennelli* DNA

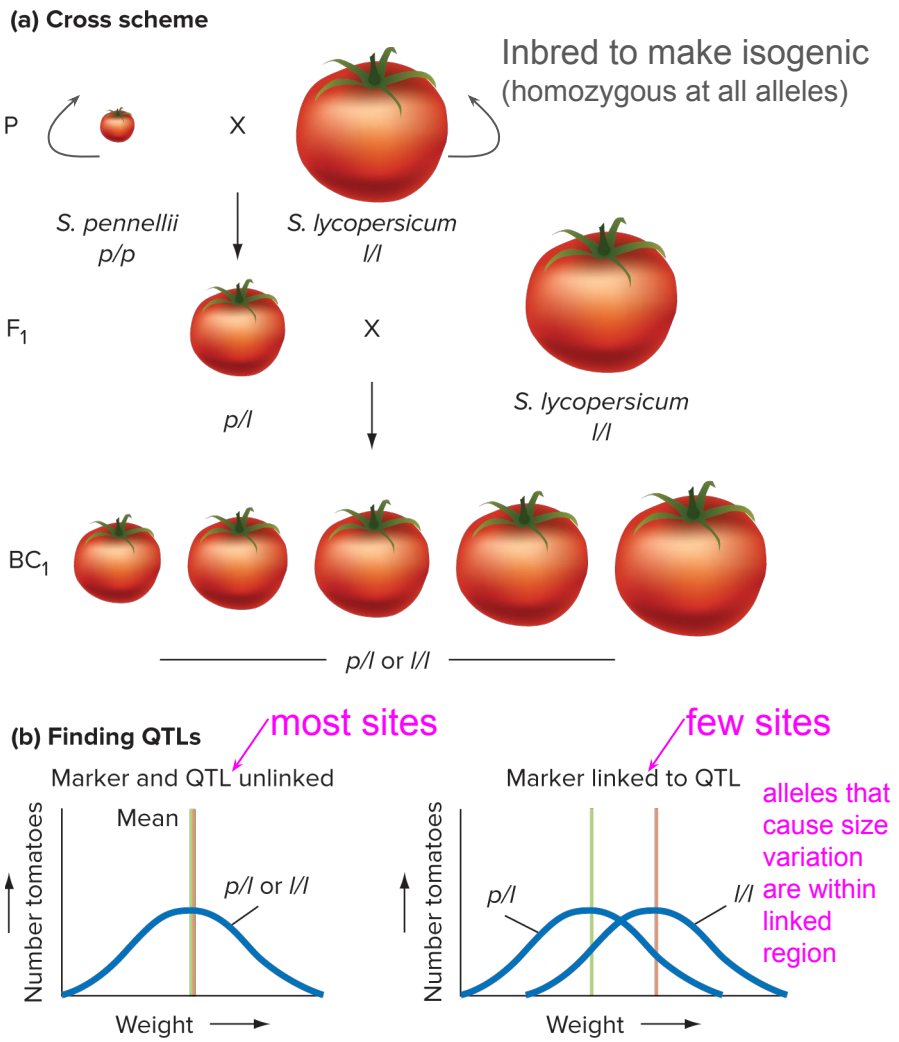
= Homozygous *Lycopersicon esculentum* DNA

= Interval in chromosome where crossover took place

a = Significantly different (P<0.01) from M82-1-8 large-fruited control.

b = Significantly different (P<0.01) from NIL 939-2 small-fruited control.

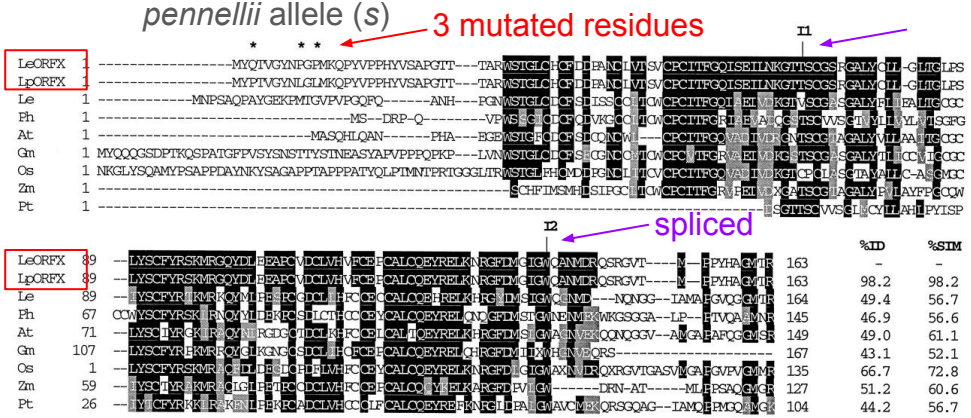
Alpert and Tanksley 1996



Fine mapping of a tomato size QTL

Wild-type fw2.2 is a negative regulator of cell division

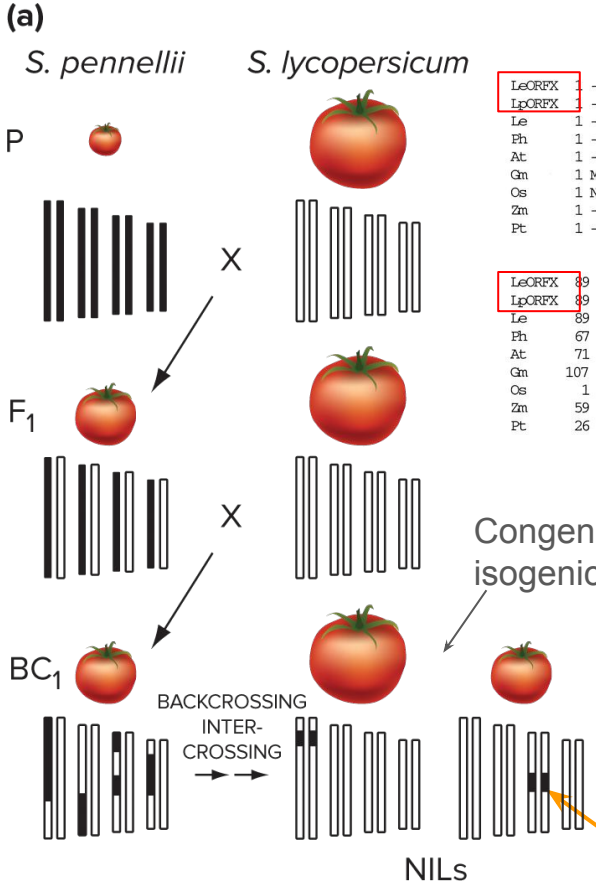
The *S. lycopersicum* allele (*l*) is hypomorphic to the *S. pennellii* allele (*s*)



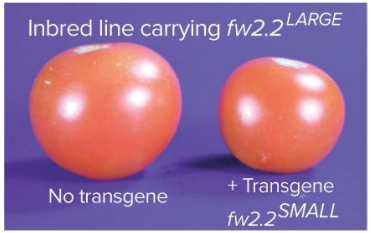
Frary et al. 2000

Isogenic lines

Backcross with *S. lycopersicum*



Congenic or nearly isogenic lines (NILs)



phenotypic rescue to validate

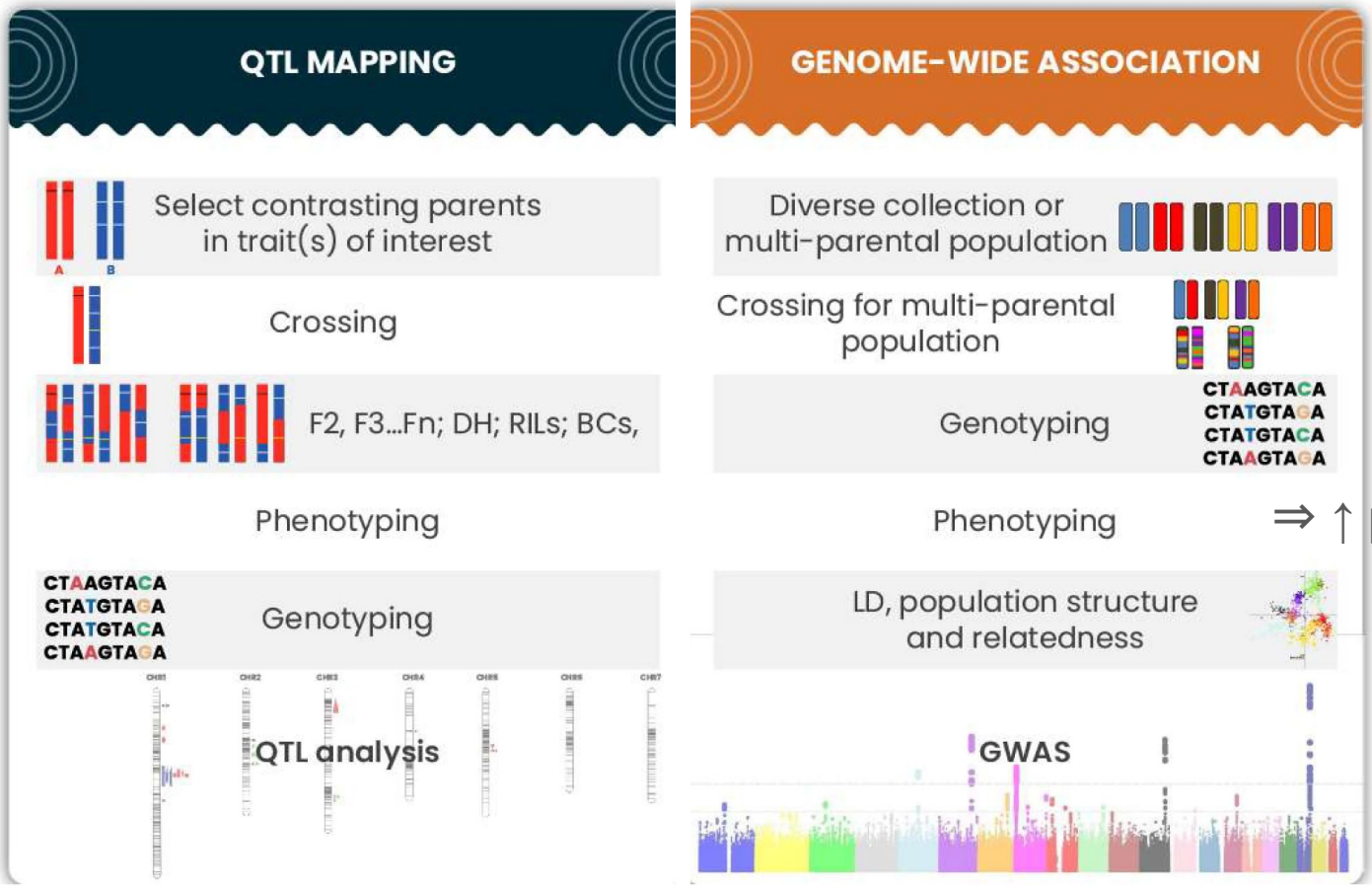
sequence to identify gene/region

Introgressed *S. pennellii* haplotype explaining ~30% weight

Phenotype → Genotype mapping

↓ alleles
↓ linkage

⇒ ↑ bp resolution



↑ alleles
↑ linkage

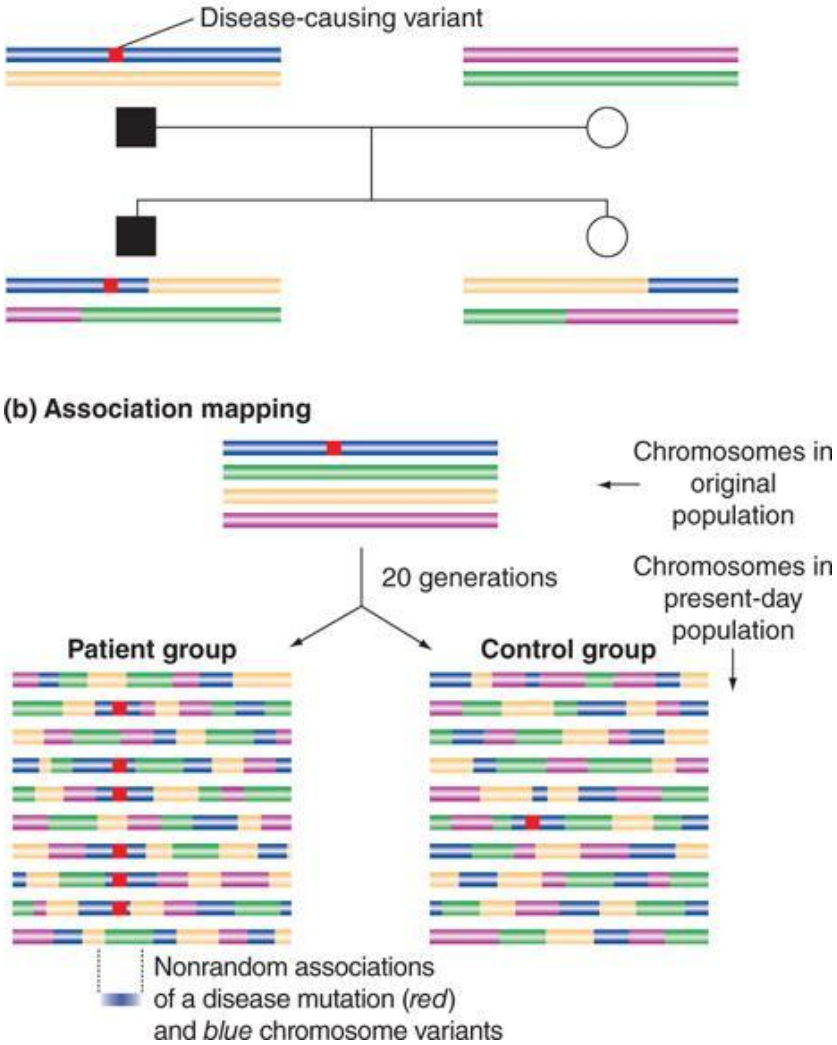
⇒ ↑ phenotypes

Riaz et al. 2021

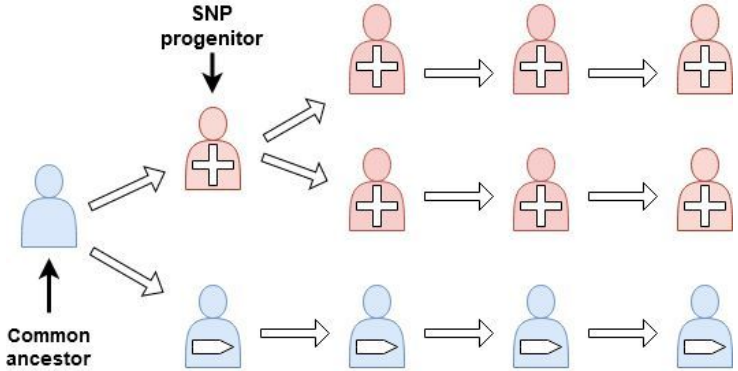
Phenotype-genotype association mapping

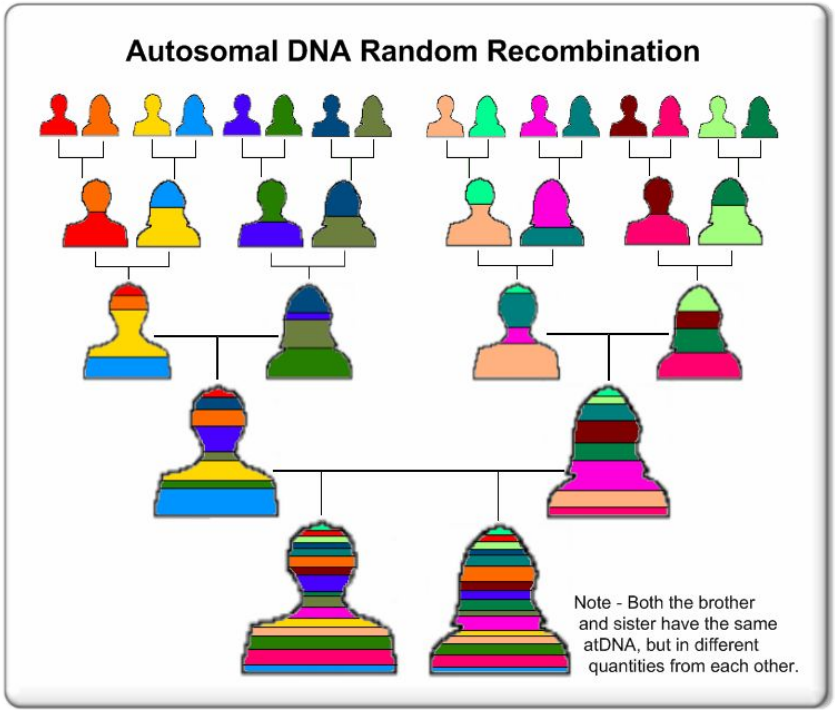
- Over many generations, recombination causes shuffling of variants that are linked on the same chromosome

- Remember that variants that are closer together are less likely to have recombination occur between them.



Phenotype-genotype association mapping





"all chromosomes in the present-day population are patchworks of the chromosomal types that were present in the population's ancestors"

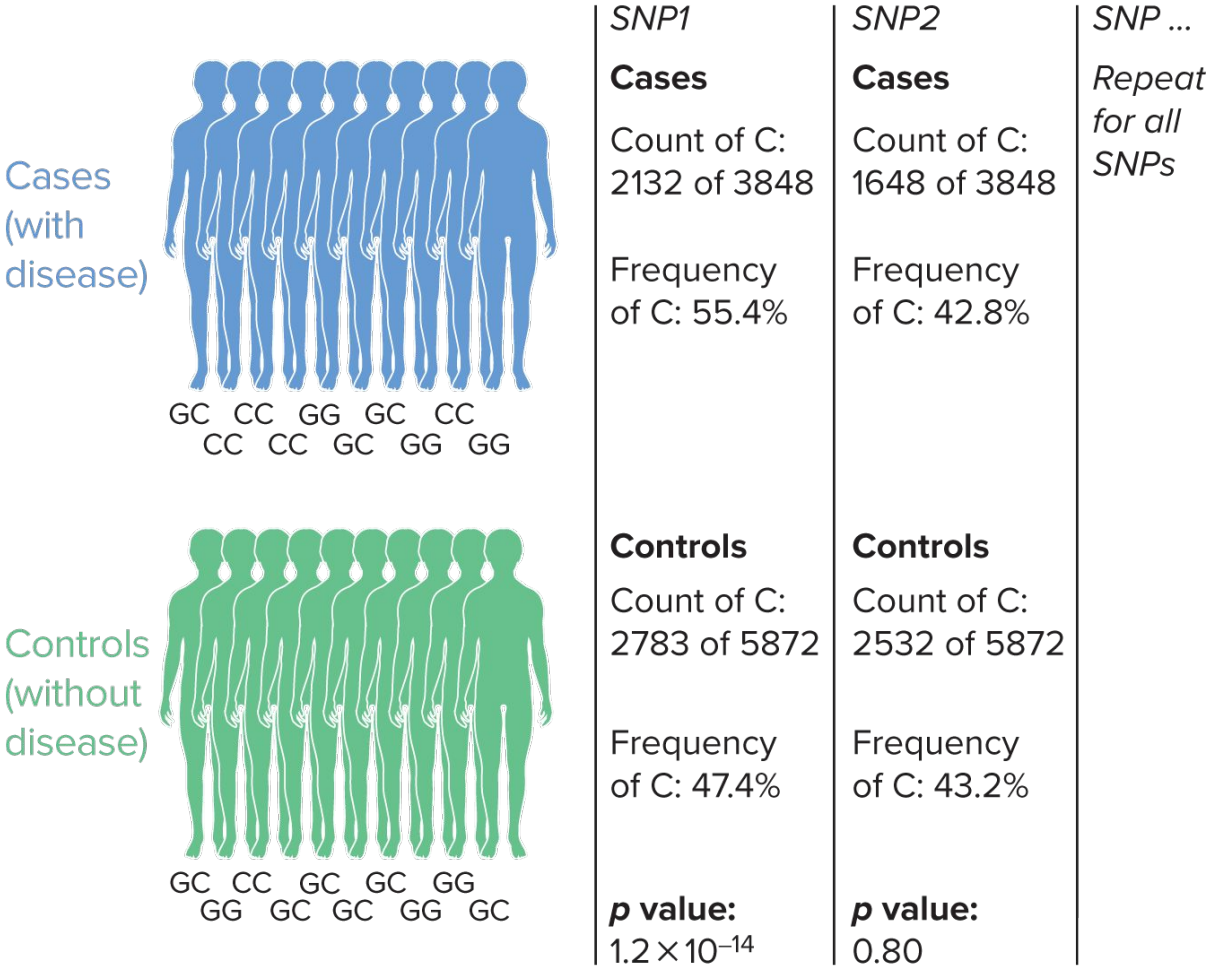
Testing for association of SNPs to cases (phenotypes)

GWAS ~ Genome-wide association studies

Test: Are any SNPs positively correlated with the trait/disease?

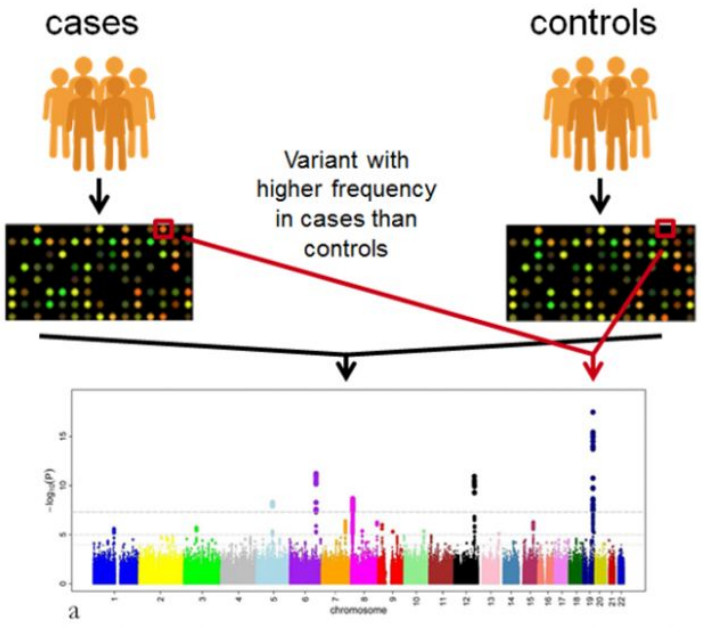
> Necessary because SNP transmission probabilities cannot be tracked/predicted like Mendelian traits

Null hypothesis: SNPs are equally distributed between cases and controls



Testing for association of SNPs to cases (phenotypes)

- The null hypothesis is that there is no association of a SNP to cases compared to controls
- We can use a statistical test to reject the null hypothesis.
- A p-value is the probability of seeing this extreme of a bias, by random chance, of a SNP to cases when the null hypothesis is true
- We typically reject the null hypothesis when $p < 0.05$.
  - Cut-off indicates level of tolerance for incorrect rejection of the null
  - This number must get smaller, as more SNPs (hypotheses) are tested
- For this class, we are using a chi-squared test for the association of the SNP to cases compared to controls. The actual methods used for GWAS are more complicated.



Contingency table for chi-squared test for independence

SNP allele	Cases	Controls	
G	$\mathbf{O_{CaG}}$ = Cases with G $\mathbf{E_{CaG}}$ = (Total G/Total genomes) x Total Cases $\mathbf{X^2_{CaG}} = (E_{CaG} - O_{CaG})^2 / E_{CaG}$	$\mathbf{O_{CoG}}$ = Controls with G $\mathbf{E_{CoG}}$ = (Total G/Total genomes) x Total Controls = Total G - E_{CaG} $\mathbf{X^2_{CoG}} = (E_{CoG} - O_{CoG})^2 / E_{CoG}$	Total G = $O_{CaG} + O_{CoG}$
C	$\mathbf{O_{CaC}}$ = Cases with C $\mathbf{E_{CaC}}$ = (Total C/Total genomes) x Total Cases = Total Cases - E_{CaG} $\mathbf{X^2_{CaC}} = (E_{CaC} - O_{CaC})^2 / E_{CaC}$	$\mathbf{O_{CoC}}$ = Controls with C $\mathbf{E_{CoC}}$ = (Total C/Total genomes) x Total Controls = Total C - E_{CaC} $\mathbf{X^2_{CoC}} = (E_{CoC} - O_{CoC})^2 / E_{CoC}$	Total C = $O_{CaC} + O_{CoC}$
	Total Cases = $O_{CaG} + O_{CaC}$	Total Controls = $O_{CoG} + O_{CoC}$	Total genomes = $O_{CaG} + O_{CoG} + O_{CaC} + O_{CoC}$

O - observed
E - expected

$$\textbf{Total } \chi^2 = \chi^2_{CaG} + \chi^2_{CaC} + \chi^2_{CoG} + \chi^2_{CoC}$$

Degrees of freedom (df)=1.
If **one** value is known, the rest of the expected values can be determined

Why is it one degree of freedom (df)?

	Category A		Total
Category B	?		6
			15
Total	10	11	21

The row and column totals are a given.

When you calculate one of the values in the blank cells, you can fill out the rest of the table. This is why the degree of freedom is 1.

Degrees of freedom = the number of <u>independently variable factors</u> affecting the range of states in which a system may exist

E.g., Finding one expected SNP count allows the rest to be calculated

Getting a p-value from a X^2 value and degrees of freedom

From a table

Percentage Points of the Chi-Square Distribution

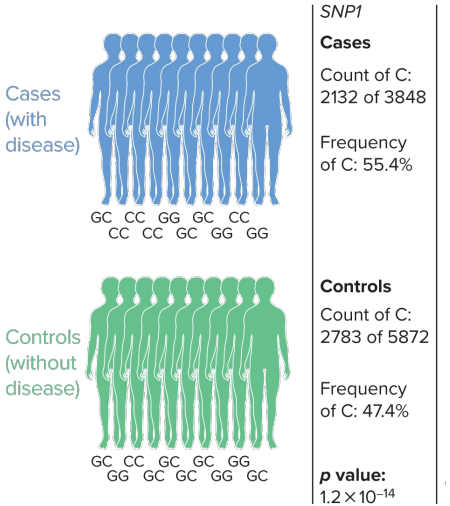
Degrees of Freedom	Probability of a larger value of x^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09

From Excel or Google Sheets

- CHIDIST(X^2 value, df)
  - Make sure the value that is returned is a scientific number.
    - In Google Sheets: Format -> Number -> Scientific

Example chi-squared test for independence

1) Calculate total G, C, genomes, cases, and controls
2) Calculate expected values of cases and controls for each SNP
3) Calculate χ^2 for cases and controls for each SNP
4) Calculate total χ^2 and find p-value from Chi-squared distribution

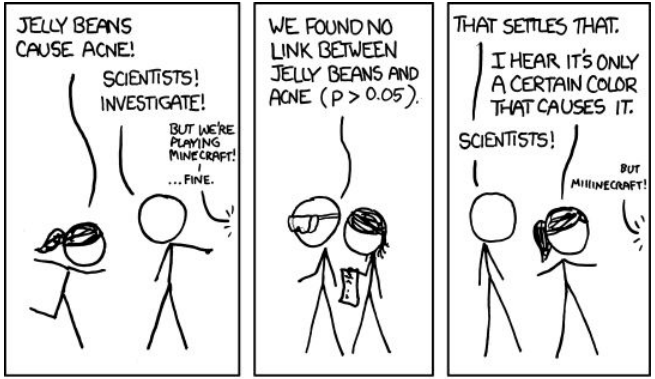


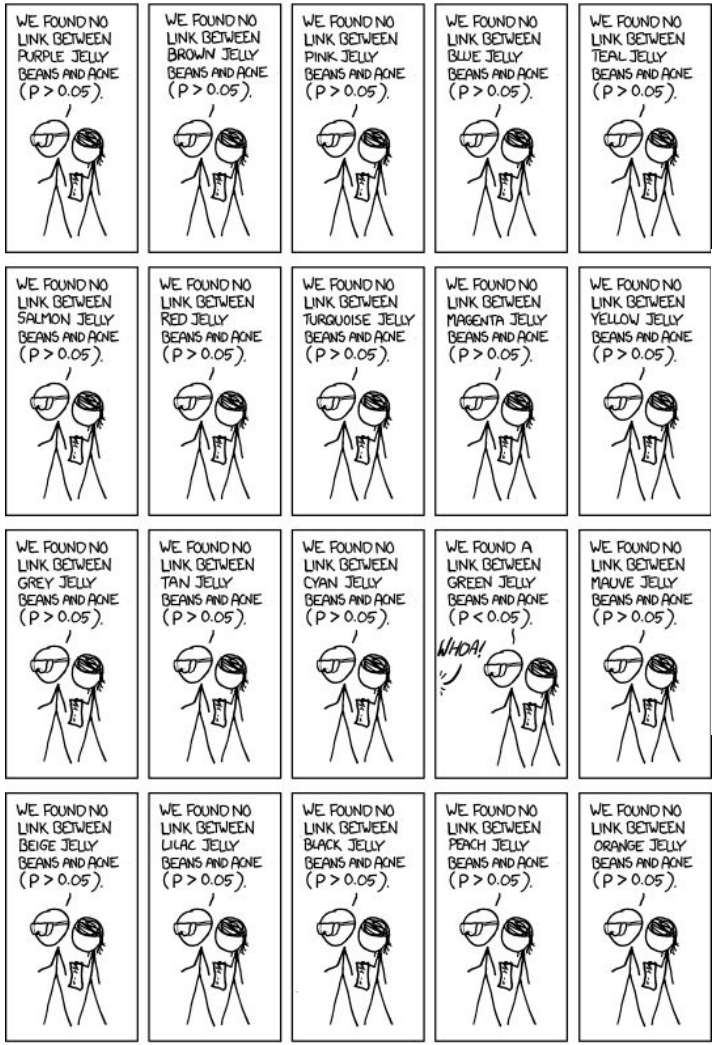
SNP1	Cases	Controls	
G	**O = 1716** **E** = [(4805/9720)] × 3848 = **1902** χ^2 = [(1716 − 1902)^2/1902] = **18.2**	**O = 3089** **E** = [(4805/9720)] × 5872 = **2903** χ^2 = [(3089 − 2903)^2/2903] = **11.9**	**Total G = 4805**
C	**O = 2132** **E** = [(4915/9720)] × 3848 = **1946** χ^2 = [(2132 − 1946)^2/1946] = **17.8**	**O = 2783** **E** = [(4915/9720)] × 5872 = **2969** χ^2 = [(2783 − 2969)^2/2969] = **11.7**	**Total C = 4915**
	Total Cases = 3848	**Total Controls = 5872**	**Total genomes = 9720**

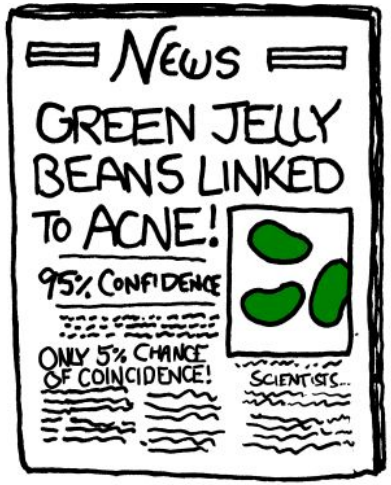
$\chi^2 = 18.2 + 11.9 + 17.8 + 11.7 = \mathbf{59.6}$; **df = 1**; $p = \mathbf{1.2 \times 10^{-14}}$

Interpretation: 1/1.2e-14 → a one in 8.33e13 chance this association happened by chance

Multiple testing must be corrected for







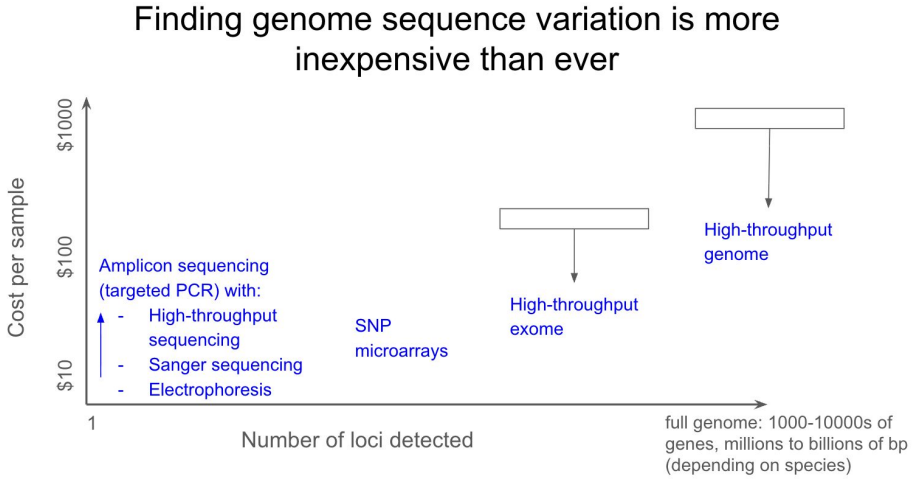
But to find significant SNP associations, we need to account for the fact that we tested every SNP.

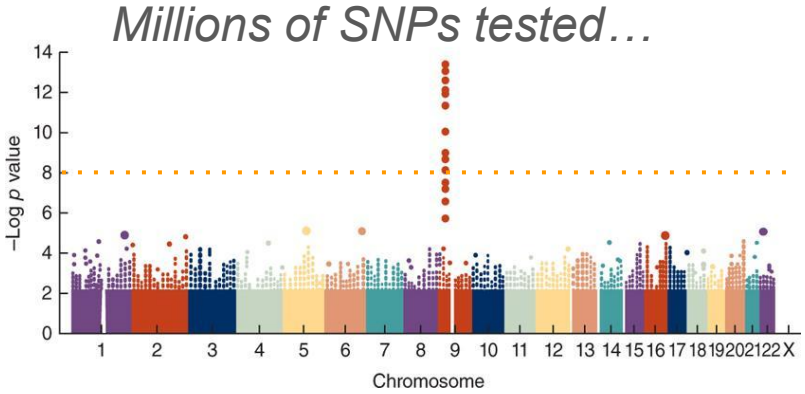
Is the association of eating green jelly bean and acne found just because you tested all possible colors?

By random chance, you expect there to be an association.

Overall significance must consider number of SNPs tested

To reject the null hypothesis, the p-value from the chi-squared test has to be less than: **<u>0.05</u>**/Number of SNPs tested





p-value = 0.05 →→→→→→→→→→ 5×10^{-8}

What is the average increase in disease risk associated with a given SNP?

(a) Calculating the allelic odds ratio

Odds of C allele given Case status = Freq of C in Case/Freq of G in Case

Odds of C allele given Control status = Freq of C in Control/Freq of G in Control

$$\text{Odds Ratio} = \frac{\text{Case C/Case G}}{\text{Control C/Control G}} = \frac{2132/1716}{2783/3089} = 1.38$$

Interpretation: Case group has 1.38 times odds of having the disease/phenotype than the control group

QTL mapping in humans often uses linear regression

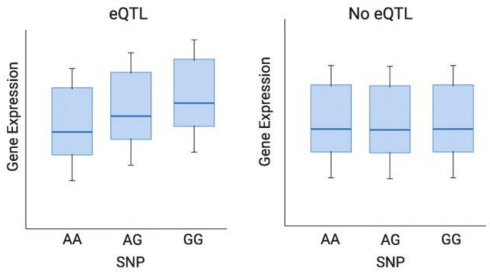
Article | Open access | Published: 30 April 2024

Genome-wide analysis in over 1 million individuals of European ancestry yields improved polygenic risk scores for blood pressure traits

Jacob M. Keaton, Zoha Kamali, Tian Xie, Ahmad Vaez, Ariel Williams, Slavina B. Goleva, Evangelos Evangelou, Jacklyn N. Hellwege, Loic Yengo, William J. Young, Matthew Traylor, Zhili Zheng, Jian Zeng, Daniel I. Chasman, Andrew P. Morris, Mark J. Caulfield, Shih-Jen H Jaspal S. Kooner, David Conen, John R. Attia, Alanna C. Morrison, Ruth J. F. Loos, Million V Program, Lifelines Cohort Study, CHARGE consortium, ICBP Consortium, ... Helen R. Warre

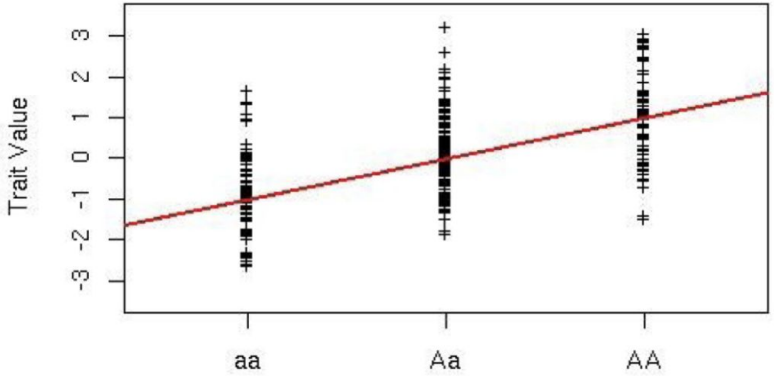
+ Show authors

Nature Genetics (2024) | Cite this article



Unrelated Samples

$$\hat{y}_i = \mu + \hat{\beta} x_i$$



Slide modified from Angela Brooks and Timothy Thornton

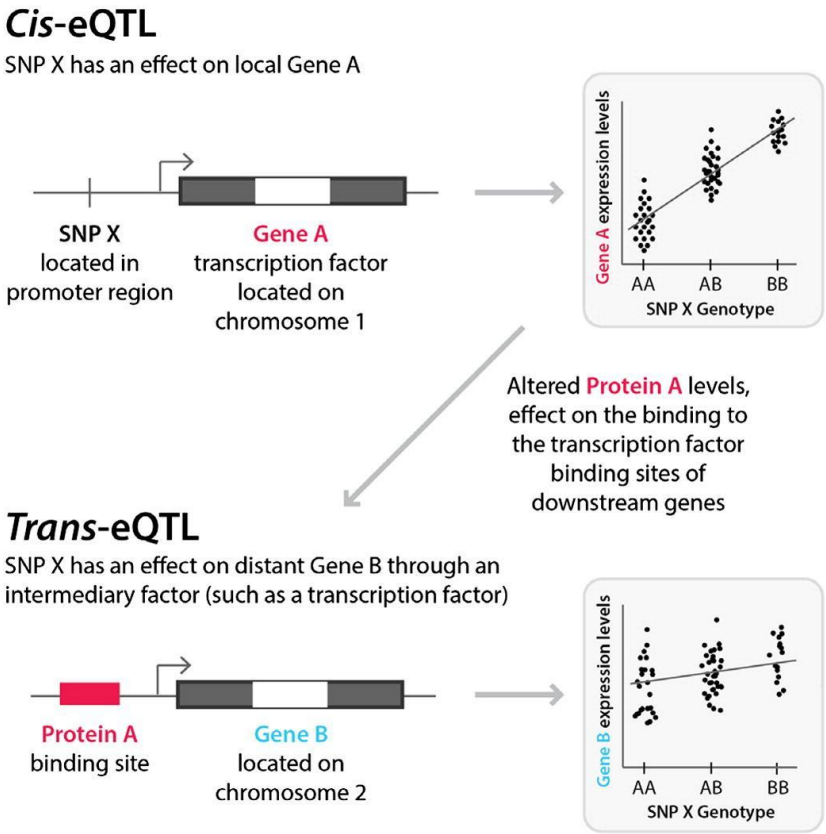
Association mapping of QTLs: expression as a phenotype

eQTL: expression QTL

eQTL analysis complements GWAS studies in humans because gene expression is a quantitative value

It is thought that a phenotypic change will likely be caused by underlying changes in gene regulation

There is a difference between *cis*- and *trans*-eQTLs

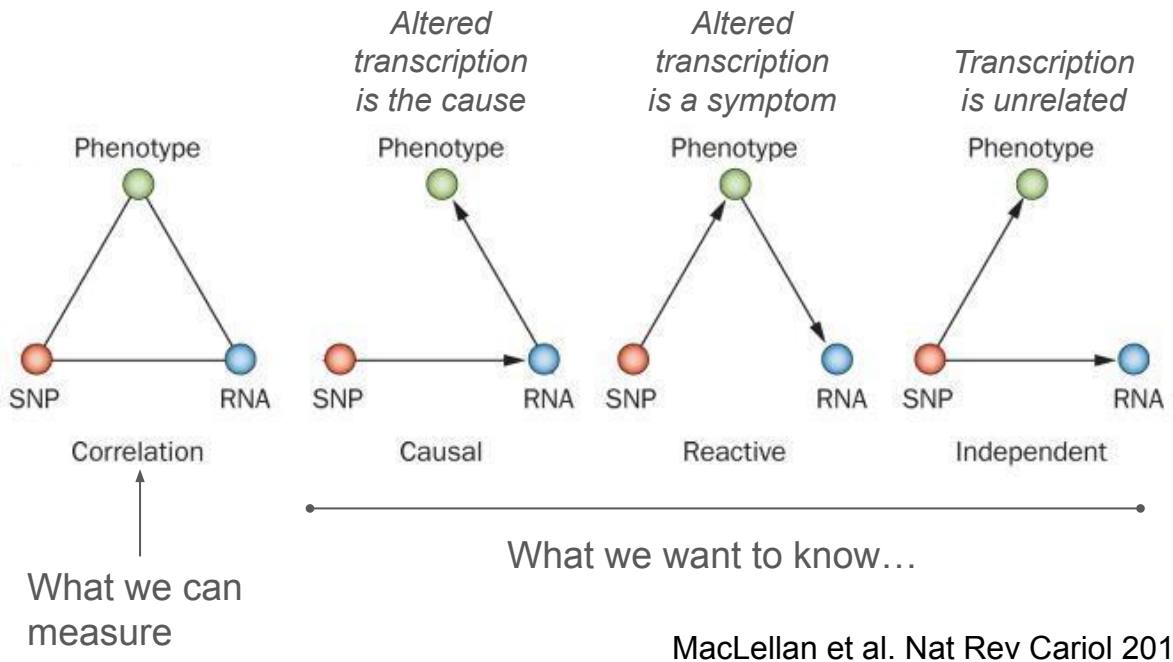


Westra and Franke 2014 BBA

eQTL analysis complements GWAS (low-res)

The association of a SNP to gene expression might point to a cause of the phenotype.

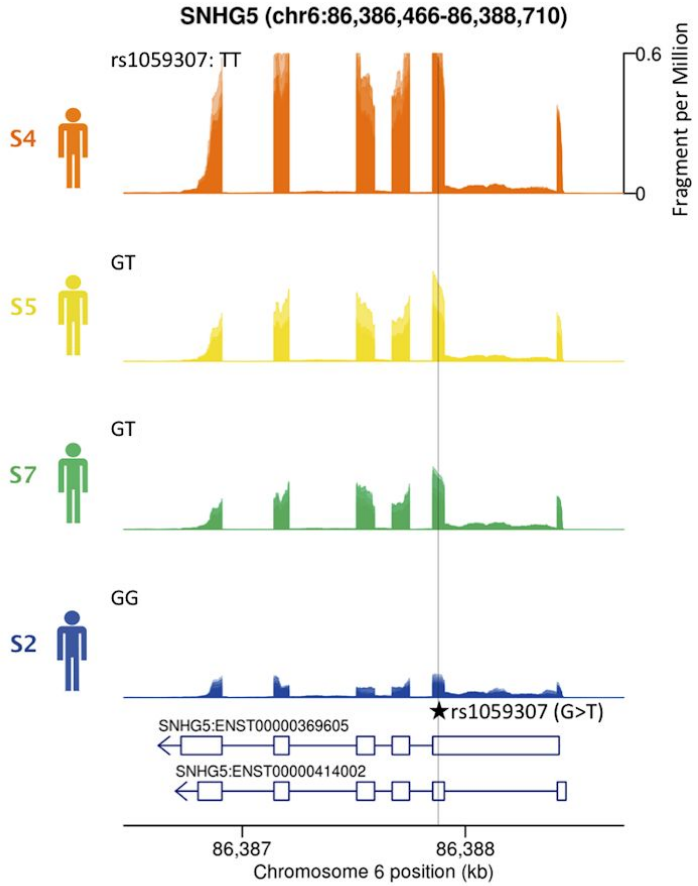
However, the gene expression change could not be directly caused by the SNP (reactive) or be independently associated with the phenotype.

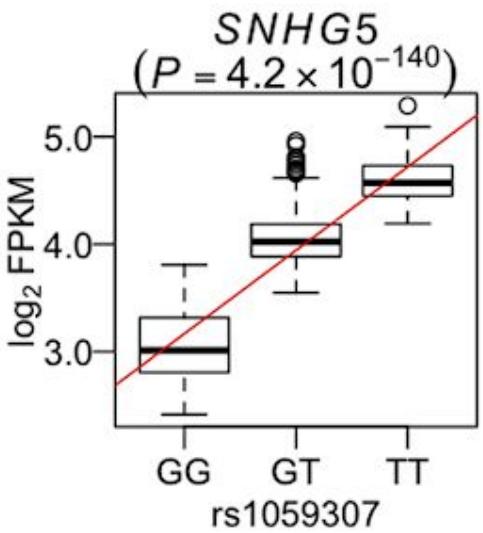


MacLellan et al. Nat Rev Cariol 2012

eQTL association identified through linear regression

cis-eQTL





p-value based on the null hypothesis that the slope of the red line is 0.

p-value is probability of observing a negative or positive slope that deviates from 0 more than you would expect by chance

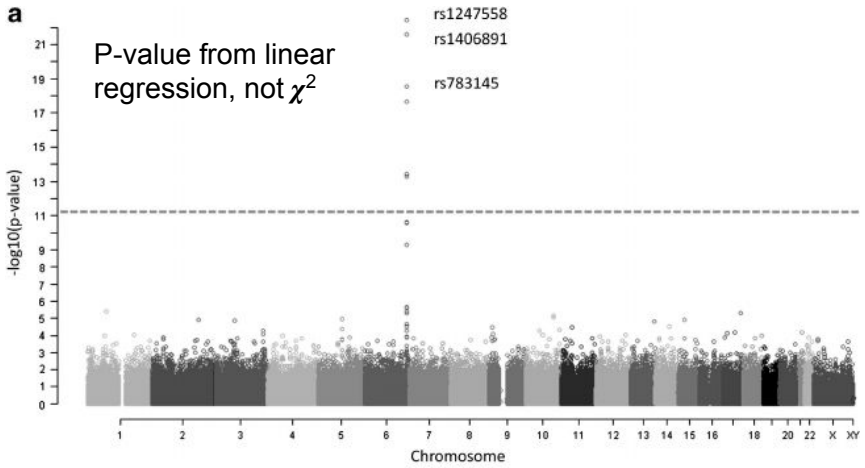
You still need to consider the multiple testing problem.

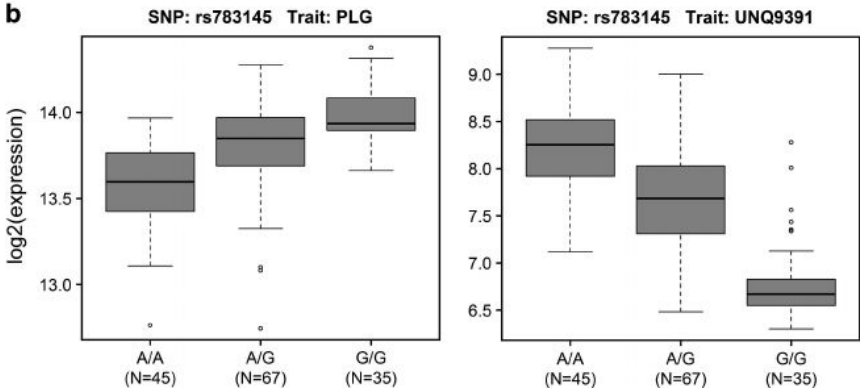
cis- and *trans*-eQTL association identified through linear regression

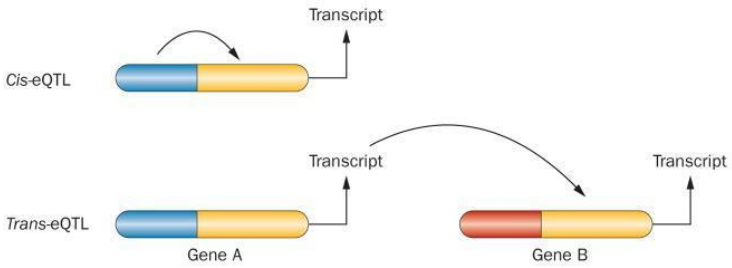
The SNP rs783145 is both a *cis*- and *trans*-eQTL.

rs783145 is located in the intron of the *PLG* gene on chr6. It is a *cis*-eQTL of *PLG*.

The SNP is also associated with altered expression of the gene *UNQ9391* on chr8. It is a *trans*-eQTL of *UNQ9391*.

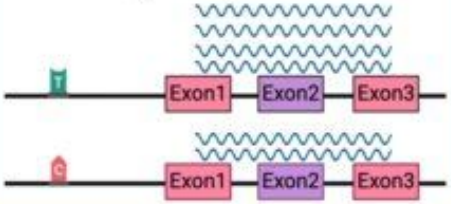




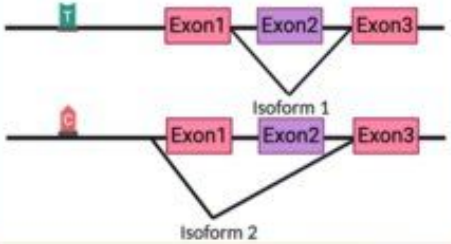


Many epigenetic QTL types

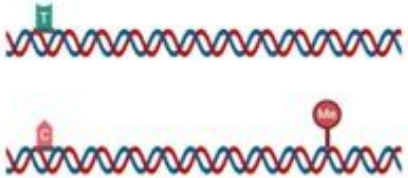
eQTL: A genetic variant that explains variation in gene expression levels



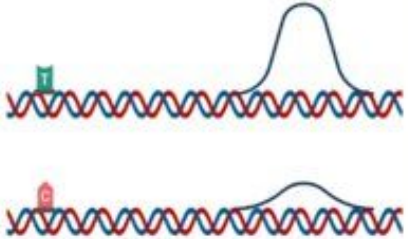
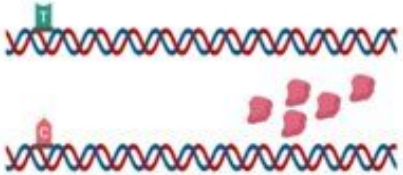
sQTL: A genetic variant responsible for variation in mRNA splicing



meQTL: A genetic variant that affects DNA methylation



caQTL: A genetic variant affecting chromatin accessibility



proQTL: A genetic variant perturbing protein expression levels